Route-2 Assault XLII S7.10: prime_weight_bounds Theorem (Khat Specialization) and the Prime-Weight Bound True Proof

1. Task and Conclusion

This round demotes the prime-weight lemma prime_weight_bounds (formerly an axiom) to a true theorem prime_weight_bounds_Khat, and specializes the main assembly lemma prime_sum_bound to K := Khat. Full lake build QRH passes: 2704 jobs, exit=0, no warnings. The boundedness/smoothness of the prime weights S(X) is now closed as a Lean true proof; the remaining premises are only the three named axioms of connector, smoothness and integral-level narrow support (Section 5).

2. Mathematical Setup

2.1 Prime sum and weights

Prime sum (the genuine prime-term carrier of BGS Lemma 3 in the difference-type pairing, S7.9 support-truncation convention):

$$S(X) = \Sigma_{1 \le n \le \lfloor X \rfloor} \Lambda(n)\chi_5(n)\cdot n^{-1/2}\cdot \hat{K}\ (\delta\cdot \log n)$$

Weight $b_n = n^{-1/2}\cdot \hat{K}\ (\delta\cdot \log n)$. $\hat{K}$ = fourierCos Khat is the cosine Fourier transform of the optimal kernel $f_0(x) = \cos(\sqrt{2}x)/(\sqrt{2}\cdot\sin(1/\sqrt{2}))$ (support $[-1/2,1/2]$), closed form:

$$\hat{K}\ (u) = [\ \sin((\sqrt{2}-2\pi u)/2)/(\sqrt{2}-2\pi u) + \sin((\sqrt{2}+2\pi u)/2)/(\sqrt{2}+2\pi u)\ ]\ /\ (\sqrt{2}\cdot\sin(1/\sqrt{2}))$$

2.2 Key judgement: the global bound

The earlier fractional bound Khat_abs_le has denominator $|\sqrt{2}\pm 2\pi u|$, which vanishes at $u = \pm\sqrt{2}/(2\pi)$ — it is not a global uniform bound and yields no constant. The correct tool is $|\sin t| \le |t|$: each sin(t)/(2t)-type factor of $\hat{K}$ satisfies $|\sin(t)/(2t)| \le 1/2$, the two terms sum to ≤ 1, hence the global uniform bound

$$|\hat{K}\ (u)| \le B = 1/(\sqrt{2}\cdot|\sin(1/\sqrt{2})|)$$

(khat_global_bound true proof; B is the constant of the boundedness branch.)

2.3 Smoothness branch decomposition

Let $A = (\sqrt{(n+1)})^{-1}$, $C = (\sqrt{n})^{-1}$. The weight difference splits as:

$$\Delta b_n = A\cdot(\hat{K}\ (\delta\cdot\log(n+1)) - \hat{K}\ (\delta\cdot\log n)) + (A-C)\cdot\hat{K}\ (\delta\cdot\log n)$$

Four true-proof lemmas (Main chain): (i) $|\log(n+1)-\log n| \le 1/n$ (log monotonic + log $x \le x-1$); (ii) $A \le C$ (inverse-sqrt monotonic); (iii) $C \cdot C = n^{-1}$ (square identity); (iv) $|A-C| \le C^3$ (rationalization $\sqrt{(n+1)}-\sqrt{n} = 1/(\sqrt{(n+1)}+\sqrt{n})$, denominator $\ge (\sqrt{n})^3$).

Smoothness: $\hat{K}$ is globally Lipschitz (band-limited entire function, supp $\hat{K} \subset [-1,1]$); there is L with $|\hat{K}(u)-\hat{K}(v)| \le L \cdot |u-v|$; with (i) this gives $|\hat{K}(\delta \cdot \log(n+1)) - \hat{K}(\delta \cdot \log n)| \le L \cdot \delta/n$. Assembly:

$$|\Delta b_n| \le A \cdot (L \cdot \delta/n) + C^3 \cdot B \le C \cdot (L \cdot \delta \cdot C \cdot C) + C^3 \cdot B = C^3 \cdot (L \cdot \delta) + C^3 \cdot B = M \cdot C^3,\ M = L \cdot \delta + B$$

2.4 Boundedness branch

By 2.2: $|b_n| \le (\sqrt{n})^{-1} \cdot |\hat{K}(\delta \cdot \log n)| \le (\sqrt{n})^{-1} \cdot B \le M \cdot (\sqrt{n})^{-1}$ ($M \ge B$).

3. Lean Implementation

3.1 Core (S7_1_Core.lean, §3/§4b)

New true proofs: abs_sin_div_two_mul_le_half (t=0 branch by Lean division-by-zero convention, subst t; simp; t≠0 branch via Real.abs_sin_le_abs), khat_global_bound (hpos positive denominator + hnum two terms ≤ 1/2 + hden via abs_mul). New named axioms: fourierCos_Khat (connector tying the abstract operator fourierCos to the Khat closed form — the reason the generic prime_weight_bounds K cannot advance and K := Khat specialization is mandatory), khat_lipschitz (smoothness, ∃L globally Lipschitz).

3.2 Main (S7_1_Main.lean)

True-proof chain: log_succ_sub_le_inv, inv_sqrt_succ_le, inv_sqrt_sq_eq_inv, inv_sqrt_succ_sub_le (items (i)–(iv) of 2.3). prime_weight_bounds_Khat assembles into a theorem (bounded + smooth branches). prime_sum_bound is specialized: signature drops the generic K and hKs; the body rcases prime_weight_bounds_Khat δ hδ; the narrow-support piece bgs5_khat_fourier_supp is moved above primeSum so it is available before prime_sum_bound.

3.3 Pitfalls logged

(1) Idempotent-replacement damage: a string replacement re-replaced the embedded fourierCos K prefix inside fourierCos Khat, producing Khathat (2 sites: the prime_sum_bound goal line and the bgs5 definition) — patterns contained in each other need ordering or a single rewrite. (2) le_add_of_nonneg_left gives a ≤ b+a (adds on the left); the goal a ≤ a+b needs le_add_of_nonneg_right. (3) Goals with ⁻¹ products need field_simp first;

plain ring is insufficient. (4) Two consecutive /-- --/ doc comments before a declaration cause a syntax error (unexpected token); the redundant comment was removed.

4. Build Evidence

lake build QRH: Core 14s ✔ → Main 11s ✔ → Stage3 22s ✔ → QRH 9.6s ✔, 2704 jobs, exit=0, no warnings. Also cleaned 6 historical unused-variable warnings (T/Z/hn/hδ/hc′/hX/hT/hε0 renamed with _ prefix; positional arguments unchanged). Debug probe Dbg_Probe.lean removed; Dbg_Khat.lean kept (khat true-proof development archive).

5. Remaining Axioms and Next Step

Three named axioms remain: fourierCos_Khat (connector), khat_lipschitz (smoothness), bgs5_khat_fourier_supp (integral-level narrow frequency support). Next: the constant L in khat_lipschitz is computable from the closed form (K̂ band-limited entire), provable by a derivative piece (same technique as khat_second_order_bound: closed-form K̂ ′ plus interval bounds), to be merged into the S7.1 Stage 3+ strike list.